

Entregáveis

Vídeo de apresentação: <https://youtu.be/qTcYB2xWUaU>

Link do repositório GitHub:

https://github.com/mario-jiroe/tech_challenge02

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Wine Quality

Tech Challenge | Pós-Tech | Data Analytics | Fase 2

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Classificando a qualidade de vinhos com Machine Learning

01.

Objetivo

Desenvolver um modelo de classificação capaz de prever a qualidade final do vinho com base nas características físico-químicas.

02.

Base de dados

Wine Quality Dataset - Kaggle

03.

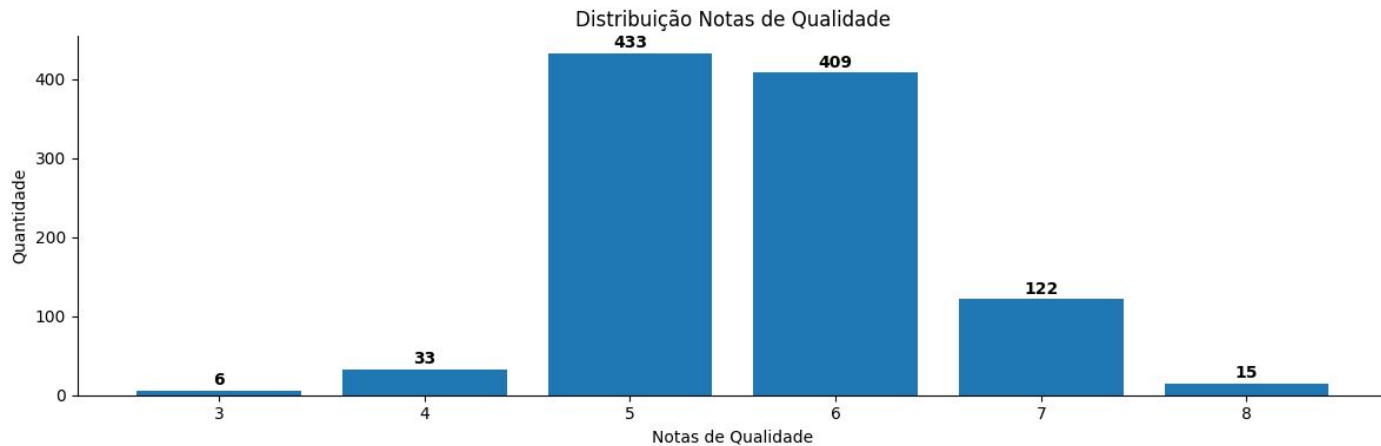
Variável determinante

Vinho de Alta Qualidade: **nota** ≥ 7

Vinho de Baixa/Média Qualidade: **nota** < 7



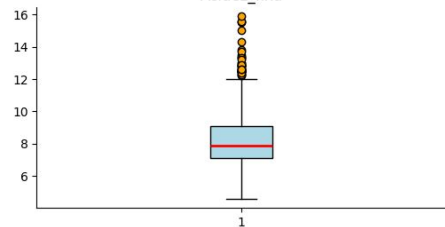
Distribuição dos dados



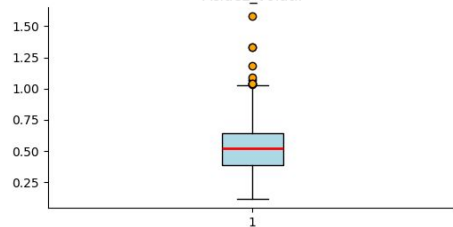
Qualidade	Acidez Fixa	Acidez volatil	Ácido Cítrico	Açúcar Residual	Cloreto	SO ₂ Livre	SO ₂ Total	Densidade	pH	Sulfatos	Teor Alcoólico
3	7.45	0.8175	0.05	2.1	0.0800	5.5	15.0	0.997565	3.355	0.56	9.825
4	7.60	0.6750	0.07	2.1	0.0780	12.0	32.0	0.996640	3.380	0.56	10.000
5	7.80	0.5900	0.23	2.2	0.0820	15.0	48.0	0.996900	3.300	0.58	9.700
6	7.90	0.5000	0.26	2.2	0.0780	14.0	35.0	0.996520	3.320	0.64	10.500
7	8.55	0.3600	0.40	2.3	0.0705	12.0	27.5	0.995495	3.275	0.74	11.600
8	8.60	0.3600	0.45	2.0	0.0720	7.0	19.0	0.995160	3.230	0.74	11.800

Análise exploratória

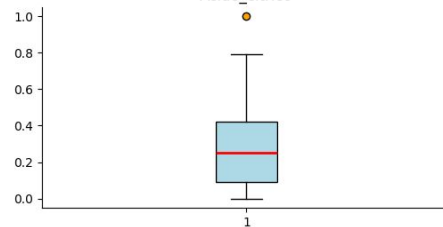
Acidez_fixa



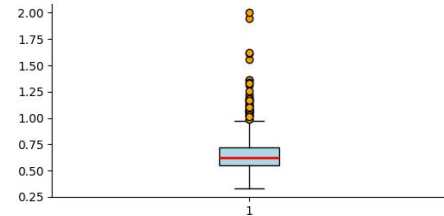
Acidez_volatil



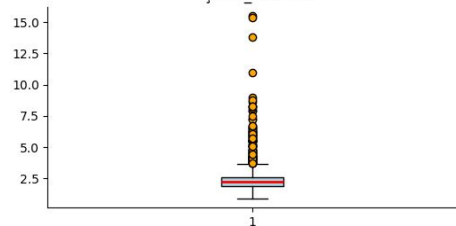
Acido_citrico



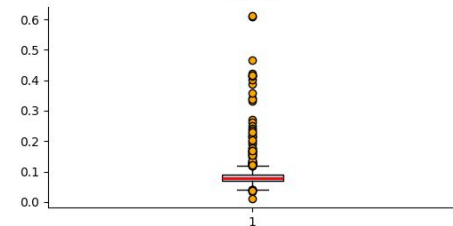
Sulfatos



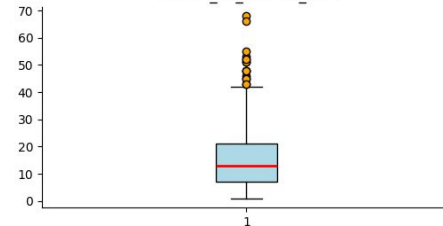
Açúcar_residual



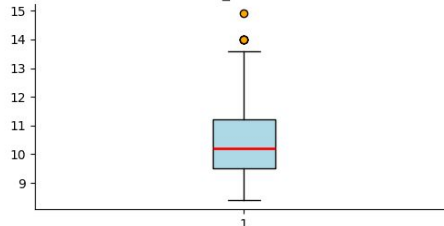
Cloretos



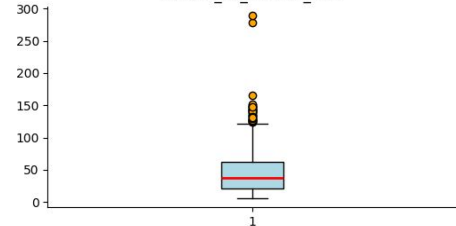
Dioxido_de_enxofre_livre



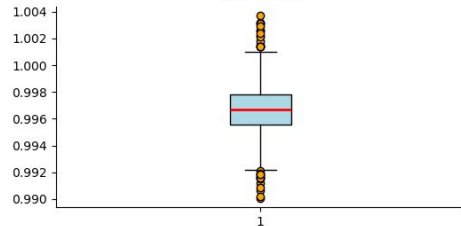
Teor_alcoolico



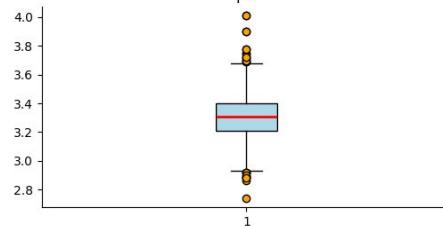
Dioxido_de_enxofre_total

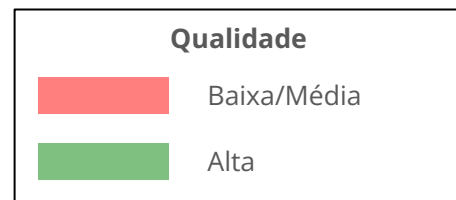
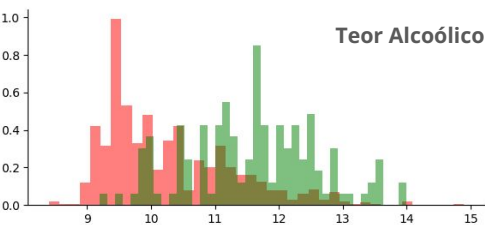
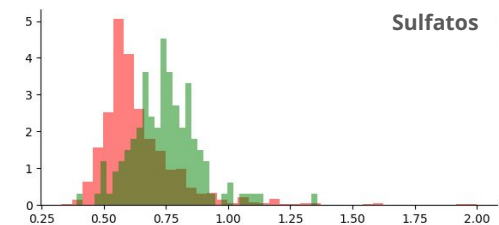
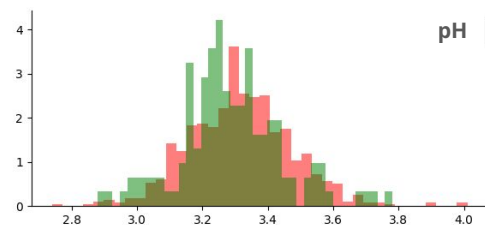
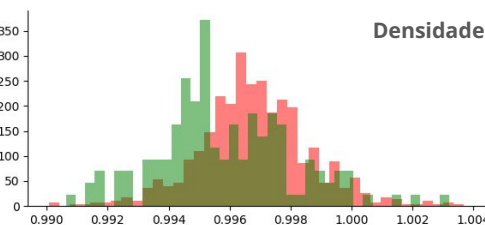
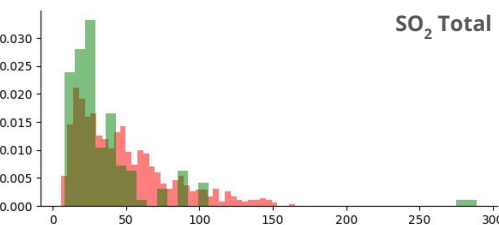
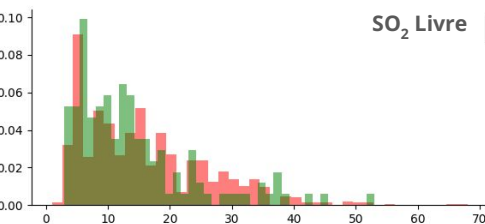
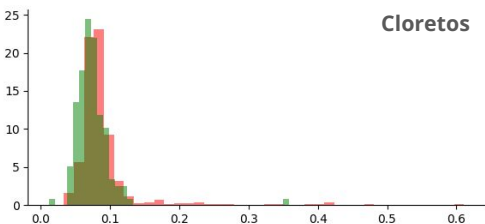
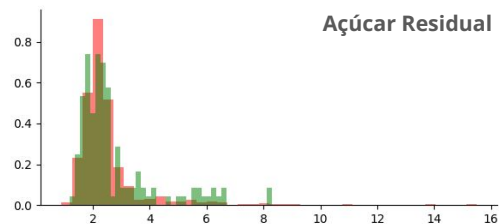
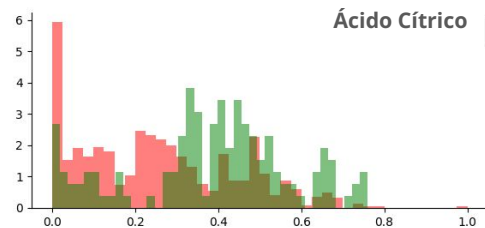
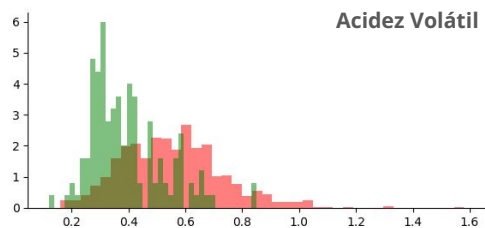
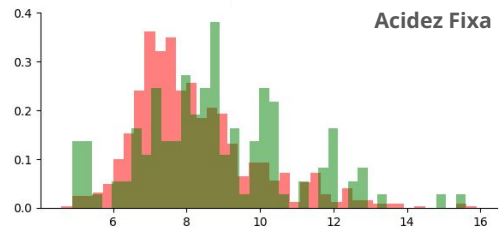


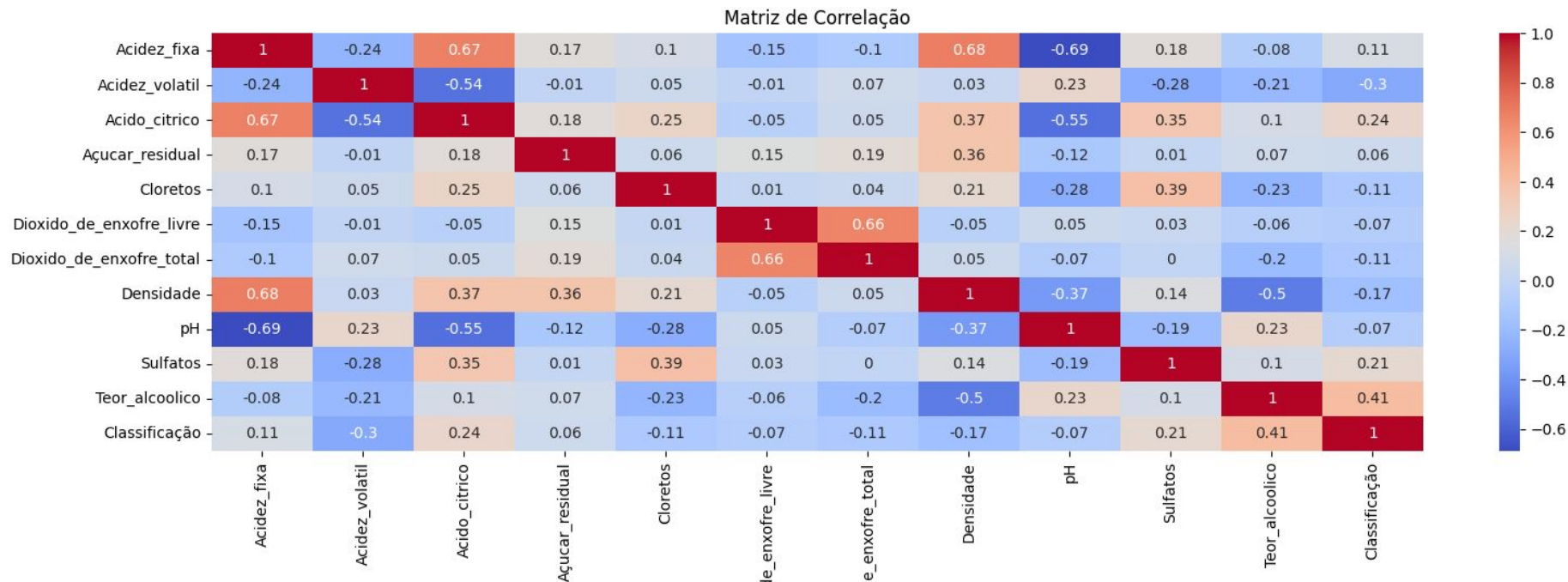
Densidade



pH









Modelos

Random Forest

Baseado em árvore de
decisão

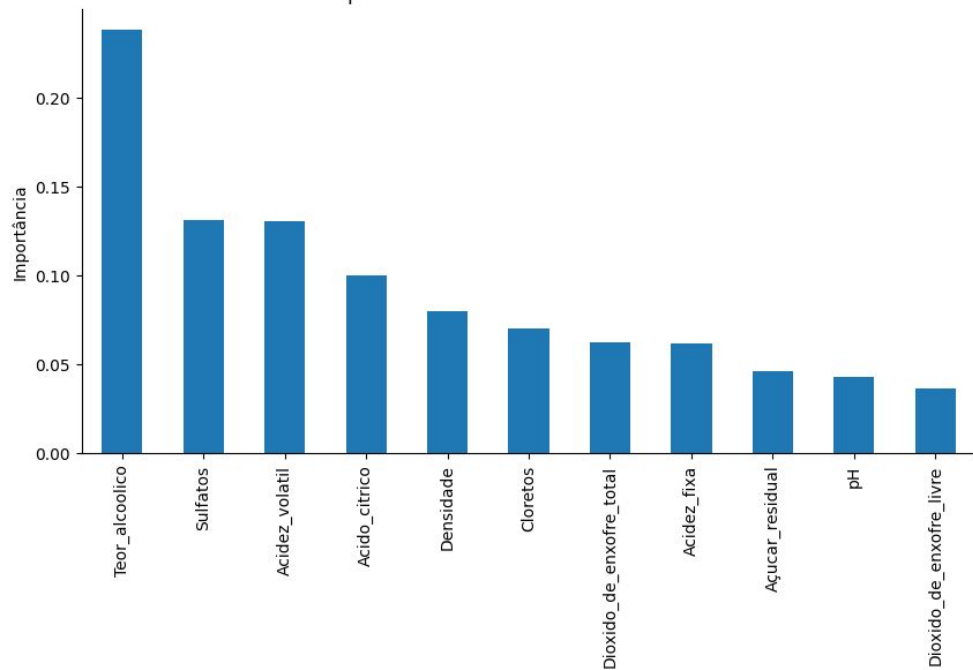
KNN

Baseado em distância
para os vizinhos próximos

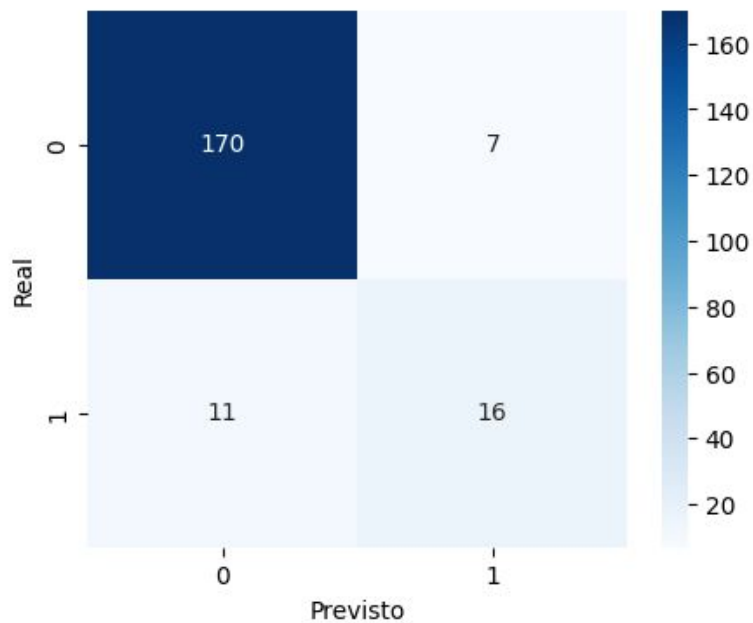


Random Forest

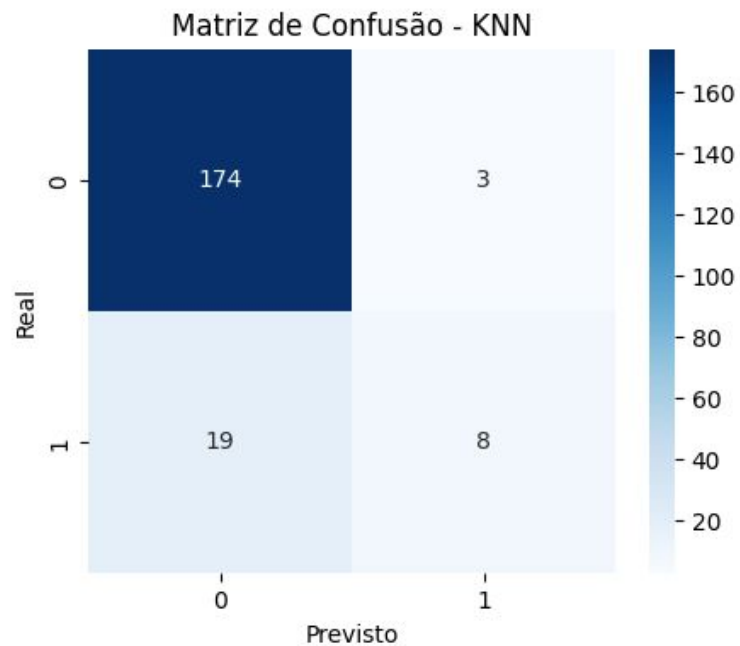
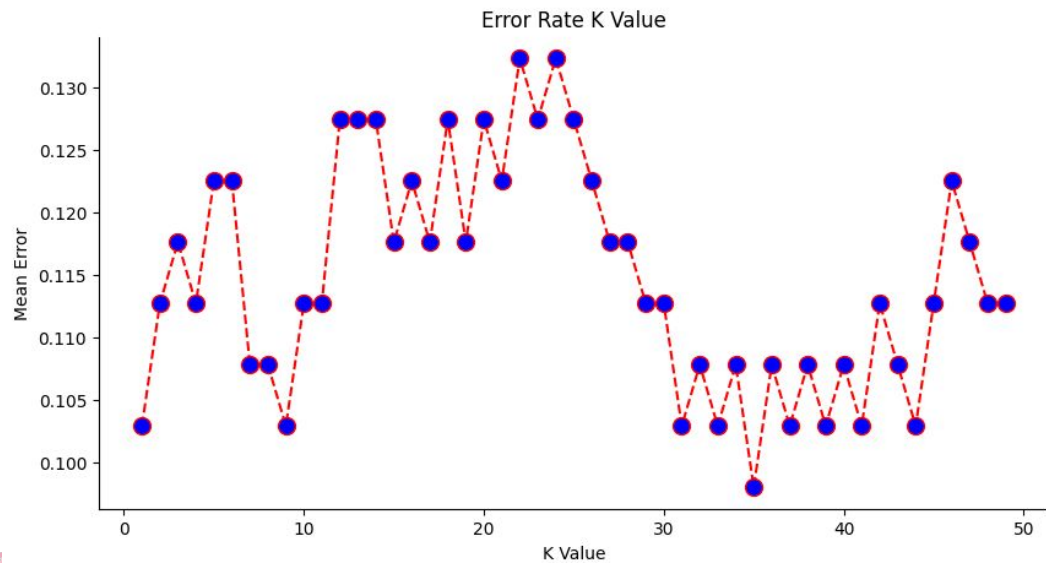
Importância das Variáveis - Random Forest



Matriz de Confusão - Random Forest



K-Nearest Neighbors



Comparativo

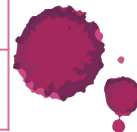


0 Baixa/Média qualidade



1 Alta qualidade

Parâmetros	K-Nearest Neighbors (KNN)	Random Forest
Acurácia	89,22%	91,18%
Recall	30%	59%
Precisão	73%	70%
F1-Score	42%	64%



Random Forest

Estrutura

Assimetria da variável-alvo



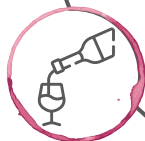
59%

Vinhos de maior qualidade



F1-Score

64% (RF) x 42% (KNN)



Precisão

73% (KNN) x 70% (RF)



Referências

ER, Yeşim; ATASOY, Ayten. The classification of white wine and red wine according to their physicochemical qualities.

International Journal of Intelligent Systems and Applications in Engineering, v. 4, n. Special Issue-1, p. 23-26, 2016.

M Yasser H. Wine Quality Dataset. Disponível em:

<https://www.kaggle.com/datasets/yasserh/wine-quality-dataset>

